Supplementary Information
The Resolution Calibration Principle: Certified Locality for Kernel and Gibbs Fields

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Abstract

This document collects the full proofs of the propositions, theorem, and corollary stated in the main paper. Each Supplementary Note states the relevant result and gives its complete derivation; the main paper carries only the statements and short proof sketches. Several Notes also contain supplementary-only propositions and corollaries, labelled with an S prefix. Equation and symbol conventions are those of the main text: a kernel field $F = (z_i, v_i)$ with Gaussian kernel $K_\sigma(t) = e^{-t/2\sigma^2}$, far set $\mathcal{F} = \{i : \|y - z_i\| > d\}$, effective mass $A = V_{\max} N_{\text{eff}}$ with participation ratio $N_{\text{eff}} = (\sum_{i\in\mathcal{F}} \omega_i)^2 / \sum_{i\in\mathcal{F}} \omega_i^2$, and certified frontier $\sigma^* = d/\sqrt{2\ln(A/\varepsilon)}$ under assumptions (A1) bounded weights $V_{\max} = 1$ held fixed as σ varies (fixed amplitude), (A2) a meaningful near/far radius d (the certificate is vacuous with margin ε when the far set is empty), and (A3) budget below mass $0 < \varepsilon < A$. The decision certificate is stated in vector ℓ_1 form for one-hot weights; a normalised (Gibbs) certificate with certified temperature τ_{cert} extends the principle to prototype and attention readouts, and a Rényi family and split-conformal wrapper are given as supplementary results.

Roadmap

This Supplementary Information is organised into six Parts. **Part A** (Notes 1–4) proves the additive certificate: the safety frontier and aggregate tail bound, the Rényi/Hill effective-mass family, one-shot conservativeness, and the frozen-geometry corollary. **Part B** (Notes 5–7) derives its operational consequences: decision stability and the performance floor, the uniform-over-region certificate, and the selective-prediction operating point. **Part C** (Notes 8–12) extends the principle to normalised (Gibbs) readouts: attention as a Gaussian kernel field, prototypical networks, the Gibbs resolution certificate (Theorem 2) with its full task-grounded HotpotQA program, output locality, and the single-head-to-full-layer bound. **Part D** (Note 13) gives the conformal finite-sample query certificate. **Part E** (Notes 14–15) reports the empirical program across vision, text and tabular fields and the hard-truncation comparator. **Part F** (Notes 16–18) covers independent numerical verification, robustness, and reproducibility. Supplementary-only results carry an S prefix (Propositions S1–S5, Corollaries S1–S2); results shared with the main text keep their main-text numbers.

Part A · The additive certificate

Part A · Note 1 — Safety frontier and aggregate tail bound

Proposition 1 (Safety frontier)**.** *For fixed $A > 0$, $d > 0$, and tolerance $0 < \varepsilon < A$, the equation* $\text{cert}(A, \sigma, d) = A\, K_\sigma(d^2) = \varepsilon$ *has the unique solution*

$$\sigma^*(A, \varepsilon, d) = \frac{d}{\sqrt{2\ln(A/\varepsilon)}}, \tag{1}$$

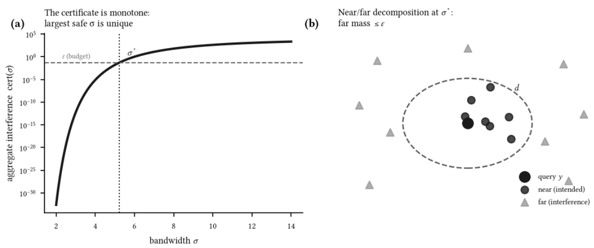

Figure S1: **The certificate.** (a) Aggregate interference $\mathrm{cert}(\sigma)$ is strictly increasing in σ; the largest σ holding it below the budget ε is the unique frontier σ^* (Proposition 1). (b) At a query, sources split into near (intended) and far (interference); σ^* bounds the far mass by ε.

and the safe set $\{\sigma > 0 : \mathrm{cert}(A, \sigma, d) \leq \varepsilon\}$ *is the interval* $(0, \sigma^*]$.

Proof. cert is continuous and strictly increasing in σ on $(0, \infty)$, from 0^+ to $A > \varepsilon$; solving $A\, e^{-d^2/2\sigma^2} = \varepsilon$ for σ gives the stated σ^*, and monotonicity gives the interval. $\square$

Aggregate tail bound. The single-source frontier extends to the whole far tail. Every far source satisfies $\|y - z_i\| > d$, so by monotonicity of the Gaussian its weight obeys $\omega_i = K_\sigma(\|y - z_i\|^2) \leq K_\sigma(d^2)$. Hence $\sum_{i \in \mathcal{F}} \omega_i^2 \leq K_\sigma(d^2) \sum_{i \in \mathcal{F}} \omega_i$, which rearranges to $\sum_{i \in \mathcal{F}} \omega_i \leq N_{\mathrm{eff}}\, K_\sigma(d^2)$. Therefore

$$\mathrm{Tail}_\sigma(y, d) = \sum_{i \in \mathcal{F}} |v_i|\, \omega_i \leq V_{\max}\, N_{\mathrm{eff}}\, K_\sigma(d^2) = \mathrm{cert}(A, \sigma, d). \tag{2}$$

with $A = V_{\max} N_{\mathrm{eff}}$, using $|v_i| \leq V_{\max}$ then the rearranged inequality. The only assumption is $\omega_i \leq K_\sigma(d^2)$ for far sources, which the near/far partition (A2) guarantees by construction. Because the far set is defined by $r_i > d$, the bound is sharp in the limit as all far-source radii decrease to d from above: there $\omega_i \to K_\sigma(d^2)$ for every $i \in \mathcal{F}$, both inequalities approach equalities, and the aggregate tail approaches $\mathrm{cert}(A, \sigma, d)$. Under the alternative convention $r_i \geq d$, equality is attained exactly at the boundary. Setting $\sigma = \sigma^*$ makes this aggregate leakage at most ε.

Part A · Note 2 — The Rényi/Hill effective-mass family

This note generalises the effective mass of Note 1 to the Rényi/Hill family (Proposition S1).

Statement. The aggregate bound of Note 1 uses the participation ratio $N_{\mathrm{eff}} = (\sum \omega_i)^2 / \sum \omega_i^2$, which is the Hill number of the far weights at order $q = 2$. It is one member of a family. For $q > 1$ define the Hill number of the (nonnegative) far weights $\{\omega_i\}_{i \in \mathcal{F}}$,

$$D_q = \Big(\sum_{i \in \mathcal{F}} p_i^q \Big)^{1/(1-q)}, \qquad p_i = \frac{\omega_i}{\sum_{j \in \mathcal{F}} \omega_j}, \tag{3}$$

with the limits $D_2 = N_{\mathrm{eff}}$ (participation ratio) and $D_\infty = \big(\sum_i \omega_i\big) / \max_i \omega_i$. The bound $\sum_i \omega_i \leq D_q K_\sigma(d^2)$ holds for every $q > 1$; the useful *ordered refinement relative to the participation ratio*—$D_q \leq D_2$, i.e. an equal-or-tighter certificate than Note 1—is the range $q \geq 2$, to which we restrict below.

Proposition S1 (Rényi aggregate bound). For every $q \geq 2$ the far tail obeys $\sum_{i \in \mathcal{F}} \omega_i \leq D_q\, K_\sigma(d^2)$, and the family is ordered $D_\infty \leq \cdots \leq D_q \leq \cdots \leq D_2$, so larger q gives an equal-or-tighter certificate. The tightest, D_∞, yields $\sum_i \omega_i \leq \big(\sum_i \omega_i / \max_i \omega_i\big) \max_i \omega_i$, i.e. the elementary bound $\max_i \omega_i \leq K_\sigma(d^2)$.

Proof. Every far source lies beyond d, so $\omega_i = K_\sigma(\|y - z_i\|^2) \leq K_\sigma(d^2) =: \kappa$ by monotonicity. Hence $\max_i \omega_i \leq \kappa$, and $\sum_i \omega_i = D_\infty \max_i \omega_i \leq D_\infty\, \kappa$. The ordering D_q nonincreasing in q is the standard monotonicity of Hill numbers (Rényi entropies are nonincreasing in their order), so $D_\infty \leq D_q \leq D_2$ for $q \geq 2$, and $\sum_i \omega_i \leq D_q \kappa$ for each such q follows from the $q = \infty$ case together with $D_q \geq D_\infty$. The $q = 2$ case is exactly Note 1. $\square$

Certified bandwidth and conservativeness. Setting $A_q = V_{\max} D_q$ and inverting $A_q K_\sigma(d^2) = \varepsilon$ gives a q-indexed frontier $\sigma_q^* = d/\sqrt{2\ln(A_q/\varepsilon)}$; since A_q decreases in q, σ_q^* increases in q, so higher orders certify a larger admissible bandwidth. The conservativeness argument of Note 3 applies to each D_q: the map $r_k^2 \mapsto \omega_k$ is monotone, so as σ grows the normalised weights p_i flatten in the majorisation order, and every Hill number D_q (a Schur-concave, symmetric functional of p) is nondecreasing in σ. One measurement at σ_{pair} therefore over-estimates $D_q(\sigma_q^*)$ for every q.

Empirical confirmation. On the three frozen CIFAR-100 backbones, measured over 10,000 held-out queries at $\varepsilon = 0.05$, the family refines the effective mass with the certified tail provably in budget throughout (D_2 is the smooth operating statistic, $D_\infty = \sum_i \omega_i / \max_i \omega_i$ the exact static endpoint, and intermediate q effective-mass refinements) (zero queries over ε at every q, zero monotonicity violations over a 10-point bandwidth grid): ResNet-18 $A_2 = 1.22 \times 10^4$ ($\sigma^* = 5.22$, tail 0.08ε) tightens to $A_\infty = 5.1 \times 10^3$ ($\sigma^* = 5.42$, tail 0.21ε), a 2.4× mass reduction; ResNet-50 $A_2 = 1.18 \times 10^4 \to A_\infty = 4.9 \times 10^3$ (2.4×); ViT-B/16 $A_2 = 1.67 \times 10^4 \to A_\infty = 9.8 \times 10^3$ (1.7×). The participation ratio is thus the loosest useful member. Since $D_\infty \le D_q \le D_2$ deterministically and every member is conservative, $q = \infty$ always certifies an equal-or-larger bandwidth, and a deployment that wants the tightest label-free certificate should use it. We nonetheless report D_2 as the primary mass for a concrete reason, not merely for continuity with Note 1: $D_2 = (\sum \omega)^2 / \sum \omega^2$ is a smooth, everywhere-differentiable function of the weights, whereas $D_\infty = \sum \omega / \max \omega$ has a non-differentiable max. A smooth reported mass keeps the certificate and any downstream optimisation well-behaved; D_2 supplies it while D_∞ does not. (The training experiment of Note 10 regularizes the exact far mass m_F, which is itself differentiable in the logits; it does not optimise D_2 or the D_2 bound. The relevance of q there is only that the *reported* certified fraction uses the D_2 coverage.) The Hill order is therefore an effective-mass refinement, not an independent conceptual axis: D_∞ is the exact static endpoint, D_2 the smooth operating statistic, and intermediate q interpolate; the ≈2× mass headroom between $q = 2$ and $q = \infty$ (recovering roughly half at $q = \infty$) is the price of smoothness (script `renyi_conformal.py` in the repository).

Part A · Note 3 — One-shot conservativeness

This note proves that one mass measurement is conservative (Proposition 2), including the general log-concave-tail case.

Proposition 2 (Conservativeness)**.** *The effective mass $N_{\text{eff}}(\sigma)$ is nondecreasing in σ. Consequently $\sigma^* \le \sigma_{\text{pair}}$, the mass measured at σ_{pair} satisfies $A(\sigma_{\text{pair}}) \ge A(\sigma^*)$, and the true tail at the deployed bandwidth obeys* $\text{Tail}_{\sigma^*}(y, d) \le A(\sigma_{\text{pair}})\, K_{\sigma^*}(d^2) = \varepsilon$.

Proof. The far set $\mathcal{F}$ is fixed by the geometric radius d and does not depend on σ, so we may differentiate over a fixed index set. Write the far kernel weights as $w_i = \exp(-r_i^2 u)$, $i \in \mathcal{F}$, with $u = 1/(2\sigma^2)$, so $N_{\text{eff}} = S_1^2/S_2$ with $S_1 = \sum_{i\in\mathcal{F}} w_i > 0$, $S_2 = \sum_{i\in\mathcal{F}} w_i^2 > 0$, and each w_i is a strictly decreasing function of u. Differentiating with $\partial_u w_i = -r_i^2 w_i$ gives $\partial_u S_1 = -\sum_i r_i^2 w_i$ and $\partial_u S_2 = -2\sum_i r_i^2 w_i^2$, so $\partial_u N_{\text{eff}} = (2S_1/S_2^2)\, B$ with $B = S_1 \sum_i r_i^2 w_i^2 - S_2 \sum_i r_i^2 w_i$. Expanding this as a double sum over index pairs and symmetrising by averaging the (i, j) and (j, i) terms,

$$B = \tfrac{1}{2} \sum_{i,j} \left(r_j^2 - r_i^2\right) (w_j - w_i)\, w_i\, w_j. \tag{4}$$

In each summand the map $r_k^2 \mapsto w_k = e^{-r_k^2 u}$ is strictly decreasing, so $(r_j^2 - r_i^2)$ and $(w_j - w_i)$ always carry opposite signs; their product is ≤ 0, and with $w_i w_j > 0$ every summand is ≤ 0. Hence $B \le 0$, giving $\partial_u N_{\text{eff}} \le 0$; since $u = 1/(2\sigma^2)$ is decreasing in σ, N_{eff} is nondecreasing in σ.

For the ordering, note it does *not* depend on the measured value of A. The frontier $\sigma^*(A) = d/\sqrt{2\ln(A/\varepsilon)}$ is a strictly decreasing function of A on $A > \varepsilon$, and $\sigma_{\text{pair}} = \sigma^*(1)$ by definition. Since the normalisation $V_{\max} = 1$ and the bound $N_{\text{eff}} \geq 1$ give $A = V_{\max} N_{\text{eff}} \geq 1$ for *any* field and *any* measurement bandwidth, we have $\sigma^*(A) \leq \sigma^*(1) = \sigma_{\text{pair}}$ before A is evaluated—the inequality is structural, not a consequence of the particular measured mass. Measuring the mass at $\sigma_{\text{pair}} \geq \sigma^*$ therefore samples N_{eff} at a larger bandwidth than deployment, and monotonicity gives $A(\sigma_{\text{pair}}) \geq A(\sigma^*)$. Substituting this larger mass into the aggregate bound $\text{Tail}_{\sigma^*}(y, d) \leq A(\sigma^*)\, K_{\sigma^*}(d^2) \leq A(\sigma_{\text{pair}})\, K_{\sigma^*}(d^2) = \varepsilon$ yields the claim; the one-shot measurement can only over-state the mass, so the certificate holds without iteration. □

The self-consistent bandwidth. The one-shot value already certifies, so no iteration is required. If one nonetheless wants the exact bandwidth at which the measurement and deployment scales coincide, note that it is the unique root of the *envelope equation* $g(\sigma) := A(\sigma)\, K_\sigma(d^2) = \varepsilon$: the mass $A(\sigma) = V_{\max} N_{\text{eff}}(\sigma)$ is nondecreasing in σ (Proposition 2) and the kernel $K_\sigma(d^2) = e^{-d^2/2\sigma^2}$ is strictly increasing in σ, so their product g is strictly increasing and crosses ε exactly once. That root is therefore located by bisection on g to any tolerance. We caution that the naive re-measurement map $A \mapsto A(\sigma^*(A))$ is *order-reversing* (a larger measured mass sets a smaller deployed bandwidth, which remeasures a smaller mass), so the usual "monotone and bounded ⇒ convergent" argument does not apply to it; the guarantee comes from the one-shot bound above, and the exact root—when wanted—from bisecting the monotone envelope. Only the one-shot value is guaranteed safe; the intermediate values of the naive re-measurement iteration need not be, since a mass measured at a smaller bandwidth need not certify the larger bandwidth it induces.

General kernels: the monotonicity is not Gaussian. The proof above used the Gaussian form $w_i = \exp(-r_i^2 u)$ only through the sign of $\partial_u w_i$; the conclusion holds for a whole class of tails. Let the kernel be $K(r/\sigma)$ for a positive, strictly decreasing K, and write $\varphi = \ln K$. For any far pair $r_i < r_j$ the weight ratio is $w_j/w_i = K(r_j/\sigma)/K(r_i/\sigma)$, and

$$\frac{d}{d\sigma} \ln \frac{w_j}{w_i} = \frac{1}{\sigma}\left[\psi(r_i/\sigma) - \psi(r_j/\sigma)\right], \qquad \psi(x) := x\,\varphi'(x) = \frac{x\,K'(x)}{K(x)}, \tag{5}$$

the *elasticity* of K. If ψ is non-increasing—equivalently, $\ln K$ is concave in $\ln x$—then $\psi(r_i/\sigma) \geq \psi(r_j/\sigma)$, so every pairwise ratio $w_j/w_i \leq 1$ rises toward unity as σ grows: the far-weight vector becomes more uniform in the majorisation order. Since $N_{\text{eff}} = (\sum w_i)^2 / \sum w_i^2$ is symmetric and Schur-concave, it is nondecreasing under this majorisation. Hence:

Proposition S2 (Conservativeness, general tail). If K is positive, strictly decreasing, and $\ln K(x)$ is concave in $\ln x$, then $N_{\text{eff}}(\sigma)$ is nondecreasing in σ, and the one-shot mass measurement is conservative exactly as in the Gaussian case. The Gaussian ($\psi = -x^2$), Laplace ($\psi = -x$), stretched-exponential ($\psi = -\frac{1}{2}x^{1/2}$), Cauchy ($\psi = -2x^2/(1 + x^2)$), and inverse-power ($\psi = -p\,x^p/(1+x^p)$) tails all satisfy the elasticity condition, and we confirm zero monotonicity violations for each on the four public datasets over a 30-point bandwidth grid. The conservativeness is therefore a property of monotone, log-concave-in-log-distance kernels, not of the Gaussian tail.

Part A · Note 4 — Frozen-geometry certificate

This note records the metric-agnostic corollary: any frozen source–query geometry inherits the certificate.

Frozen geometry. The frontier and monotonicity proofs use the source–query distances only through their fixed numerical values. Let $\rho_i(y) \geq 0$ be any source–query dissimilarities that are fixed and independent of the calibrated scale, and set $\omega_i(s) = \exp[-\rho_i(y)^2/2s^2]$ with far set $\mathcal{F}(y) = \{i : \rho_i(y) > d\}$.

Corollary S1 (Frozen-geometry certificate). For a fixed query and frozen geometry, the aggregate-envelope bound (Proposition 1) and the conservativeness of the one-shot mass measurement (Proposition 2) hold with $\rho_i(y)$ in place of the Euclidean distance $\|y-z_i\|$, because both proofs depend on the geometry only through the fixed values $\rho_i(y)$ and their ordering relative to d. In particular the certificate applies to Euclidean feature distance, a fixed global Mahalanobis metric $\rho_i(y)^2 = (y-z_i)^\top M(y-z_i)$ with $M \succ 0$ frozen, source-local quadratic forms $(y-z_i)^\top G_i(y-z_i)$ with frozen $G_i \succ 0$, and fixed graph-geodesic or diffusion dissimilarities, provided an out-of-sample query distance is defined without changing the graph during calibration. *Boundary.* If the representation, metric, graph or local covariance changes with the calibrated scale, the one-shot monotonicity argument no longer applies automatically: the geometry must be frozen, or the certificate re-computed at each new geometry. Optimising the geometry (learned anisotropic metrics, diffusion or effective-resistance distances, tangent/normal budgets) can improve the usefulness of the near/far split without altering the certificate, and is left as future work.

Part B · Operational consequences

Part B · Note 5 — Decision stability and performance floor

Vector far contribution. Let each source carry a class-value vector $v_i \in \mathbb{R}^C$ and write the far contribution as $T(y) = \sum_{i\in\mathcal{F}} v_i K_\sigma(\|y-z_i\|^2)$. The aggregate bound of Note 1 applied coordinatewise, together with $\|v_i\|_1 \le V_{\max}$, gives $\|T(y)\|_1 \le \varepsilon$ at $\sigma \le \sigma^*$. For a one-hot Parzen field each v_i is a class indicator, so $\|v_i\|_1 = V_{\max} = 1$ and this assumption holds exactly.

Theorem 1 (Decision stability, ℓ_1 form)**.** *Let $\sigma \le \sigma^*$. For the local top class a and any competitor b, the two-class gap satisfies $|T_a - T_b| \le |T_a| + |T_b| \le \|T\|_1 \le \varepsilon$. Hence a local margin $m(y) > \varepsilon$ makes the deployed prediction equal the local prediction.*

Proof. $S_a - S_b = (S_a^{\text{near}} - S_b^{\text{near}}) + (T_a - T_b) \ge m(y) - \varepsilon > 0$, so a remains the deployed argmax. □

Corollary 1 (Performance floor)**.** *At $\sigma \le \sigma^*$, deployed accuracy $\ge$ (near-field accuracy) $-$ $\Pr[\,m(y) \le \varepsilon\,]$.*

Proof. Every query with $m(y) > \varepsilon$ has deployed prediction equal to local prediction, hence is correct whenever the local prediction is. Deployed error can exceed near-field error only on $\{m(y) \le \varepsilon\}$; subtracting its measure from the near-field accuracy gives the floor. □

Corollary 2 (Far-set locality)**.** *Let $\sigma \le \sigma^*$ and let each source carry an ℓ_1-bounded value vector ($\|v_i\|_1 \le V_{\max} = 1$). Then: (i) deleting all far sources moves the output by at most ε in ℓ_1; (ii) replacing the far set by any admissible far set (one whose effective mass also satisfies the budget) shifts the far contribution by the difference of two tails, of ℓ_1-norm at most 2ε, so a near-field margin $m(y) > 2\varepsilon$ preserves the decision; (iii) arbitrary insertion of additional far sources is not covered and requires re-calibration or an independent uniform-mass bound (Note 6).*

Proof. By the aggregate bound of Note 1 applied coordinatewise with $\|v_i\|_1 \le V_{\max}$, the far contribution obeys $\|T(y)\|_1 \le V_{\max} \sum_{i\in\mathcal{F}} K_\sigma(\|y-z_i\|^2) \le \varepsilon$ at $\sigma \le \sigma^*$. Deletion sets $T(y) \to 0$, so the output moves by $\|T(y)\|_1 \le \varepsilon$, giving (i). For (ii), an admissible replacement far set $\mathcal{F}'$ has $\|T'(y)\|_1 \le \varepsilon$ by the same bound, so the shift $\|T(y) - T'(y)\|_1 \le \|T(y)\|_1 + \|T'(y)\|_1 \le 2\varepsilon$; combined with the decision-stability theorem above, a margin exceeding 2ε keeps the local argmax. Part (iii) is immediate: inserting sources can inflate the far effective mass past A, violating the hypothesis of Note 1. □

Norm-dependent thresholds. The threshold is set by the norm the value vectors obey. With one-hot (or any ℓ_1-bounded) values, $\|T\|_1 \le \varepsilon$ and the pairwise gap moves by at most ε, giving the ε threshold above. With only a per-class bound $\|v_i\|_\infty \le V_{\max}$, each class score moves by at most ε and the pairwise gap by at most 2ε, recovering the generic 2ε theorem. The main paper leads with the ℓ_1 form because the implemented one-hot field satisfies it. Admissible far-set *replacement* shifts T by the difference of two tails, of ℓ_1-norm at most 2ε, so a margin above 2ε preserves the decision under replacement. A query that fails its threshold is not misclassified—it is merely *uncertified*, and a deployer may abstain on exactly this set.

Part B · Note 6 — Uniform-over-region certificate

Proposition S3 (Uniform-over-region certificate). Fix a centre c and its far set $\mathcal{F}_c = \{i : \|c - z_i\| > d\}$, and let $\gamma_c = \min_i \big|\, \|c - z_i\| - d \,\big|$ be its margin to the threshold. For any ball radius $\delta < \gamma_c$, the near/far partition is constant on $B(c, \delta)$: no source crosses the radius anywhere in the ball, so $\mathcal{F}_y = \mathcal{F}_c$ for all $y \in B(c, \delta)$. Defining the regional tail through this single fixed set,

$$\sup_{y \in B(c,\delta)} \mathrm{Tail}_\sigma(y, d) \;\le\; \mathrm{Tail}_\sigma(c, d) + L\,\delta, \qquad L = V_{\max}\,|\mathcal{F}_c|\,\frac{d}{\sigma^2}\,K_\sigma(d^2), \tag{6}$$

provided $d \ge \sigma$, with Lipschitz constant L independent of the ambient dimension n. Choosing σ so the right-hand side is $\le \varepsilon$ certifies the whole region.

Proof. Because $\delta < \gamma_c$, every far source stays at distance $> d$ and every near source at distance $< d$ throughout $B(c, \delta)$, so $\mathrm{Tail}_\sigma(\cdot, d)$ is the sum over the *same* set $\mathcal{F}_c$ at every point of the ball. Each summand $|v_i|\,K_\sigma(\rho_i^2)$ with $\rho_i = \|y - z_i\|$ has gradient of norm $|v_i|\,(\rho_i/\sigma^2)\,K_\sigma(\rho_i^2)$. The map $\rho \mapsto (\rho/\sigma^2)e^{-\rho^2/2\sigma^2}$ is decreasing for $\rho \ge \sigma$, so for every far source, where $\rho_i > d \ge \sigma$, the per-source slope is bounded by $V_{\max}\,(d/\sigma^2)\,K_\sigma(d^2)$. Summing over the $|\mathcal{F}_c|$ far sources and applying the mean-value inequality from c to $y \in B(c, \delta)$ gives the stated L. $\square$

Why the far-set count, not the participation ratio. The Lipschitz constant sums the *gradient magnitudes* of the individual far Gaussians, and a bound on a sum of $|\mathcal{F}_c|$ per-source slopes cannot be replaced by the participation ratio N_{eff}: N_{eff} compresses the far weights through a ratio of ℓ_1 and ℓ_2 norms, an operation valid for the tail *value* (Note 1) but not for a sum of slopes, where every far source contributes its full gradient. The Lipschitz constant therefore carries the raw count $|\mathcal{F}_c|$, not N_{eff}. It remains exponentially small through $K_\sigma(d^2)$: at $\sigma = \sigma^*$, $K_{\sigma^*}(d^2) = \varepsilon/A$, so $L = (|\mathcal{F}_c|/A)(d\varepsilon/(\sigma^*)^2)$, larger than the erroneous constant when $|\mathcal{F}_c| \gg N_{\text{eff}}$ and comparable when the far set overlaps little.

Practical interpretation. On the flagship ResNet-18 field at $\varepsilon = 0.05$ the certified ball radius is $\delta \approx 0.5$ in the standardised feature metric—strictly positive, so the region certificate is non-vacuous, but small, about 1.6% of the median inter-query distance (ResNet-50 gives $\delta \approx 1.0$, also 1.6%; ViT gives $\delta \approx 0.9$, about 2.2%). The uniform-over-region certificate is therefore genuine but tight; the per-query certificate of Theorem 1, which needs no such ball, carries the main deployment claims.

Part B · Note 7 — Selective-prediction operating point

The abstention readout gives a label-free, per-query support signal: a query with no near mass carries no certificate, and the field declines it. To place its operating point quantitatively, we compare against a margin-threshold selective predictor on the same σ^* vote at the coverage the ℓ_1 certificate itself flags. At the certificate's 36% coverage on ResNet-18 a labelled margin threshold reaches 67% accuracy, against the field's full-coverage 57% (72% vs 51% at 31% coverage on ResNet-50, 87% vs 72% at 61% coverage on ViT-B/16). The certified-stable set therefore sits *on* the field's accuracy–coverage curve rather than above it: σ^*'s distinguishing property is not a higher accuracy at matched coverage but that its selection rule is label-free and carries an a priori interference guarantee, where the margin threshold needs labels to set. This is the sense in which σ^* is a single object—one bound on the far field, read three ways—rather than a competitor to labelled selective prediction.

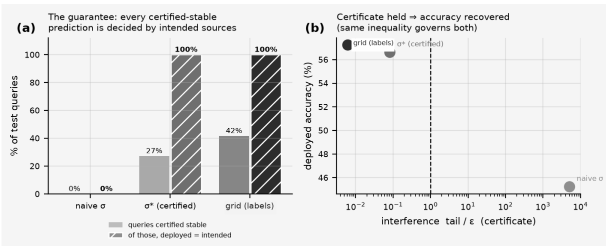

Figure S2: **The performance certificate** (frozen ResNet-18, CIFAR-100). (a) Fraction of test queries flagged certified-stable (light), and—of those—the fraction whose deployed prediction equals the intended local prediction (hatched); at σ^* and at the grid-search σ this is exactly 100% (Theorem 1). (b) The certificate and deployed accuracy move together: holding tail $\le \varepsilon$ is what recovers accuracy.

Part C · The normalised (Gibbs) extension

Part C · Note 8 — Attention as a Gaussian kernel field

This note shows softmax attention is exactly a Gaussian kernel field.

The reduction. Single-head softmax attention assigns query q the weight $a_i = \mathrm{softmax}_i(\langle q, k_i\rangle/\tau)$ over keys $\{k_i\}$. The polarisation identity $\langle q, k_i\rangle = \frac{1}{2}(\|q\|^2 + \|k_i\|^2 - \|q-k_i\|^2)$ gives

$$\exp\big(\langle q, k_i\rangle/\tau\big) = e^{\|q\|^2/2\tau} \cdot e^{\|k_i\|^2/2\tau} \cdot e^{-\|q-k_i\|^2/2\tau}. \tag{7}$$

The three factors are, in order, a query-only term $e^{\|q\|^2/2\tau}$, a per-key source weight $v_i = e^{\|k_i\|^2/2\tau}$, and the Gaussian kernel $K_\sigma(\|q-k_i\|^2)$ with $\sigma^2 = \tau$. The query-only factor is independent of i and cancels in the softmax normalisation. Hence

$$a_i = \frac{v_i\, K_\sigma(\|q-k_i\|^2)}{\sum_j v_j\, K_\sigma(\|q-k_j\|^2)}, \qquad v_i = e^{\|k_i\|^2/2\tau}, \quad \sigma = \sqrt{\tau}, \tag{8}$$

which is a normalised Gaussian kernel field with source weights v_i and bandwidth $\sigma = \sqrt{\tau}$. This representation is exact *at a fixed temperature* τ, and the source weight $v_i(\tau) = e^{\|k_i\|^2/2\tau}$ depends on that temperature.

Scope: what transfers and what does not. The temperature-dependence of v_i has a sharp consequence for which additive results carry over.

- Proposition 1 (the interference-tail bound, stated in units of $V_{\max} = \max_i v_i$) can be *evaluated at a fixed* τ: at that τ the field is an ordinary Gaussian kernel field and the tail bound holds, with $A = V_{\max} N_{\mathrm{eff}}$ measured at that temperature.
- Proposition 2 (one-shot conservativeness) does *not* transfer unchanged when key norms vary. Conservativeness assumes source amplitudes fixed as the bandwidth changes; here every v_i moves with τ, so a temperature computed by inverting the envelope at one τ is not a one-shot certificate for another. We therefore do *not* call $\tau^* = (\sigma^*)^2$ a one-shot certified temperature through Proposition 2; it is a descriptive fixed-temperature scale. The operational, temperature-explicit attention certificate is the direct Gibbs bound of Note 10, whose one-shot minimum $\tau_{\mathrm{cert}} = \min(\tau_0, \tau_{OS})$ is proved without a fixed-amplitude assumption.
- Equal-norm keys ($\|k_i\| \equiv \mathrm{const}$) are the special case in which v_i is common and cancels in the softmax, so attention is a *pure* Gaussian field and both the tail bound and the far-set locality readout (Corollary 2) apply directly.

As a fixed-temperature diagnostic, at τ^* the additive far-interference tail $\mathrm{Tail}/V_{\max}$ stays within budget, while the normalised far-attention mass—the operationally relevant quantity, governed by Note 10—is 0.35 (BERT) and 0.62 (GPT-2), not ε-small.

Empirical confirmation (generic sanity check on pretrained models). This paragraph is a reduction-and-bound sanity check on off-the-shelf pretrained models; it is *not* part of the unified task-grounded pipeline of Note 10, which uses a single continued-trained reader with an externally supplied evidence partition. On the real query/key vectors of every sampled head of pretrained BERT-base and GPT-2 over long contexts ($\sim$360 keys/head, near/far taken as the top-decile-logit versus remaining keys), the polarisation identity holds to the models' float32 precision ($< 10^{-4}$ over all heads; it is an algebraic equality, exact up to floating-point rounding), and at τ^* the certified far-key tail Tail/$V_{\max}$ stays below ε across $\sim 35{,}000$ queries per model with zero violations, rising 18–33$\times$ over budget at $4\tau^*$ (script and outputs in the repository).

Part C · Note 9 — ProtoNet: bounded weights

This note treats prototypical networks as the bounded-weight instance.

The reduction. A prototypical network assigns a query x to class k by $p(k \mid x) = \mathrm{softmax}_k(-\|f(x) - c_k\|^2/\tau)$, where f is the embedding and c_k the class prototype (the mean support embedding). This is *immediately* a Gaussian kernel field over the prototypes: with $\sigma^2 = \tau/2$,

$$p(k \mid x) = \frac{K_\sigma(\|f(x) - c_k\|^2)}{\sum_j K_\sigma(\|f(x) - c_j\|^2)}, \qquad v_k = 1,\ \sigma = \sqrt{\tau/2}. \tag{9}$$

No polarisation is needed—the readout is already a squared-distance kernel—and every source weight is $v_k = 1$, so assumption (A1) holds with $V_{\max} = 1$ *exactly*. Unlike attention, both the interference-tail certificate (Proposition 1) and the full far-set locality readout (Corollary 2), which compares the far mass to the total output, transfer with ε-tightness.

Empirical confirmation. We train a standard 4-layer convolutional ProtoNet (64-dimensional embedding) on Omniglot (20-way episodes) to 91.5 % held-out accuracy, and certify its prototype fields on 12,000 held-out queries from 60-way episodes, with d the tenth-percentile query–prototype distance and $A = N_{\mathrm{eff}}$ measured at the single-source scale σ_{pair} (as in Note 3). The identity holds to float32 ($< 10^{-6}$); at σ^* the certified far-prototype *unnormalised* tail is 0.25ε with zero violations over all 12,000 queries, rising $\sim 560\times$ over budget at $4\sigma^*$. Because the weights are bounded ($V_{\max} = 1$), the additive certificate transfers at full strength here—the interference tail *and* the full far-set locality readout (Corollary 2), which attention cannot get. The *normalised* softmax far mass at σ^* is 0.08 (against 0.35–0.62 for attention); this is not ε-small and is not the quantity the additive certificate bounds—it is a Gibbs probability mass, certified separately by the one-shot Gibbs theorem (Note 10, Theorem 2), which on the corrected experiment holds every query below ε (Fig. 5a of the main text; script and outputs in the repository).

Part C · Note 10 — Gibbs certificate and task-grounded attention

This note proves the normalised (Gibbs) far-mass certificate (Theorem 2).

Motivation. The base certificate (Notes 1 and 3) bounds the *unnormalised* far tail $\sum_{i\in\mathcal{F}} v_i\omega_i$. Several modern readouts are *normalised* Gibbs fields—softmax attention, prototypical networks, Nadaraya–Watson regression, energy-based models—where the output-relevant quantity is the normalised far mass $m_F = \sum_{i\in\mathcal{F}} p_i$ with $p_i = e^{-E_i/\tau}/\sum_j e^{-E_j/\tau}$. For these, deleting the far set changes numerator and denominator together, and the unnormalised bound does not transfer verbatim. The following certificate bounds m_F directly.

Theorem 2 (Gibbs far-mass certificate). Let $p_i = e^{-E_i/\tau}/\sum_j e^{-E_j/\tau}$ be a Gibbs field with energies E_i and a *given* partition of the keys into a nonempty near set $\mathcal{N}$ and far set $\mathcal{F}$. The bound holds for any such partition; its interpretation depends on how the partition is chosen. An *external* partition (fixed by geometry or by a task's evidence annotation, before the field is read) makes the certificate a genuine test of whether the field respects that declared locality; an *endogenous* partition read from the energy ordering itself instead reports the field's own concentration and is a diagnostic, not an external test. Let $E_{\mathcal{N}} = \min_{i\in\mathcal{N}} E_i$ be the best near key and $\Delta = \min_{i\in\mathcal{F}} E_i - E_{\mathcal{N}}$ the near/far energy gap. When $\Delta > 0$, let $D_q^{\mathcal{F}}$ be the Hill number (Note 2) of the anchor-shifted far weights $\{e^{-(E_i-E_{\mathcal{N}})/\tau}\}_{i\in\mathcal{F}}$. Here $D_q^{\mathcal{F}} = D_q^{\mathcal{F}}(\tau)$ depends on τ through the far weights. Then the normalised far mass obeys the pointwise bound

$$m_F(\tau) = \sum_{i\in\mathcal{F}} p_i \;\le\; D_q^{\mathcal{F}}(\tau)\, e^{-\Delta/\tau}. \tag{10}$$

Because $D_q^{\mathcal{F}}$ is itself temperature-dependent, the closing of this bound to an executable deployment temperature comes in two forms, exactly as for the additive σ^* (Note 3).

One-shot certificate. Measure the Hill number $D_{q,0} = D_q^{\mathcal{F}}(\tau_0)$ once at a reference temperature τ_0 (e.g. the field's deployment temperature), and set

$$\tau_{\rm OS} = \frac{\Delta}{\ln\!\big(D_{q,0}/\varepsilon\big)}, \qquad \tau_{\rm cert} = \min(\tau_0, \tau_{\rm OS}). \tag{11}$$

Then $m_F(\tau_{\rm cert}) \le \varepsilon$. *Self-consistent frontier.* Alternatively, the exact root of $D_q^{\mathcal{F}}(\tau)\, e^{-\Delta/\tau} = \varepsilon$ is unique and found by bisection.

Proof. Divide numerator and denominator of each p_i by $e^{-E_{\mathcal{N}}/\tau}$: with $\hat{E}_i = E_i - E_{\mathcal{N}}$, $p_i = e^{-\hat{E}_i/\tau}/Z$ where $Z = \sum_j e^{-\hat{E}_j/\tau} \ge e^{-\hat{E}_{\mathcal{N}^\star}/\tau} = 1$ (the best near key contributes $\hat{E} = 0$). Hence $m_F = \sum_{i\in\mathcal{F}} e^{-\hat{E}_i/\tau}/Z \le \sum_{i\in\mathcal{F}} e^{-\hat{E}_i/\tau}$. Write $w_i = e^{-\hat{E}_i/\tau}$ for $i \in \mathcal{F}$; each $\hat{E}_i \ge \Delta$, so $\max_{i\in\mathcal{F}} w_i \le e^{-\Delta/\tau}$. By Note 2 applied to the far weights, $\sum_{i\in\mathcal{F}} w_i \le D_q^{\mathcal{F}} \max_{i\in\mathcal{F}} w_i \le D_q^{\mathcal{F}} e^{-\Delta/\tau}$, giving the pointwise bound. The bound holds for any externally supplied partition with $\Delta > 0$; when the external partition instead yields $\Delta \le 0$ (some far key outranks every near key) the query-head is *uncertified under that partition* and is neither dropped nor repaired by moving the boundary.

For the one-shot form, note that warming flattens the far distribution while cooling sharpens it: for $\hat{E}_i > \hat{E}_j$ the ratio $w_i/w_j = e^{-(\hat{E}_i-\hat{E}_j)/\tau}$ moves toward unity as τ increases and toward 0 as τ decreases, so the far weights become more uniform with warming and $D_q^{\mathcal{F}}(\tau)$ is nondecreasing in τ (the majorisation/Schur-concavity argument of Note 2). Two cases. If $\tau_{\rm OS} \le \tau_0$ then $\tau_{\rm cert} = \tau_{\rm OS}$, and by flattening $D_q^{\mathcal{F}}(\tau_{\rm cert}) \le D_q^{\mathcal{F}}(\tau_0) = D_{q,0}$, so $m_F(\tau_{\rm cert}) \le D_{q,0} e^{-\Delta/\tau_{\rm OS}} = \varepsilon$ by the definition of $\tau_{\rm OS}$. If $\tau_{\rm OS} > \tau_0$ then $\tau_{\rm cert} = \tau_0$ and $D_{q,0} e^{-\Delta/\tau_0} < D_{q,0} e^{-\Delta/\tau_{\rm OS}} = \varepsilon$, so already $m_F(\tau_0) \le D_{q,0} e^{-\Delta/\tau_0} < \varepsilon$. Either way $m_F(\tau_{\rm cert}) \le \varepsilon$. For the self-consistent form, $\tau \mapsto D_q^{\mathcal{F}}(\tau) e^{-\Delta/\tau}$ is a product of two positive nondecreasing factors, hence increasing, and crosses ε once. □

Why the naive one-line τ^* is not automatic. Writing $\tau^* = \Delta/\ln(D_q^{\mathcal{F}}/\varepsilon)$ with $D_q^{\mathcal{F}}$ left unspecified is circular: the temperature one solves for is the temperature at which the mass must be measured. The one-shot fix breaks the circularity by measuring at a fixed reference τ_0 and deploying at $\min(\tau_0, \tau_{\rm OS})$; this is the direct Gibbs analogue of the additive one-shot theorem, where A is measured at $\sigma_{\rm pair}$ and $\sigma^* \le \sigma_{\rm pair}$.

Two consequences. (i) The bound has *no source-amplitude factor*: unlike the additive Gaussian view of attention (Note 8), where the source weights $v_i = e^{\|k_i\|^2/2\tau}$ are unbounded and enter A through $V_{\max}(\tau)$, the Gibbs certificate controls the output-relevant normalised mass directly. Attention, prototypical networks, and normalised kernel regression are therefore *exact* instances: for attention $E_i = -\langle q, k_i\rangle$ and Δ is the near/far logit gap; for a prototypical network $E_i = \|f(x) - c_i\|^2$ and $\Delta = d_{\rm far}^2 - r_{\rm near}^2$. (ii) It supplies the per-head certified far mass $m^h \le \varepsilon$ that the multi-head layer bound (Note 12) requires, closing that note's remaining looseness at temperatures $\tau \le \tau^*$.

Task-grounded attention (HotpotQA): protocol, audit, deletion, controls, training. The following subsections report a single unified experiment. One task-performing reader is used throughout: `bert-base-uncased` with the pre-trained `pf-hotpotqa` QA adapter, continued-trained on 1,500 retained examples (this is exactly the matched λ=0 baseline of the training subsection), giving exact-match 0.407 on held-out data. The earlier weaker frozen-adapter reader is not used. All statistics below are on this reader unless noted.

Protocol. On the HotpotQA distractor split (Yang et al., EMNLP 2018) we truncate contexts to 512 tokens and retain only examples where every annotated supporting sentence and the answer survive (21% retained; 1,437 dev examples, split 287 validation / 1,150 test; span answers only, yes/no dropped). The near set $\mathcal{N}$ is the question and annotated supporting-fact tokens (with [CLS], [SEP]); the far set $\mathcal{F}$ is the remaining distractor context, fixed before any attention is read. We call $\mathcal{F}$ *outside annotated evidence*, not provably irrelevant. Query positions are the predicted answer start; the analysed layers are the upper four, all heads. Gold supporting-fact annotations define the partition (and, in the training subsection, the penalty).

Certificate audit. The certificate is evaluated as a family of distinct quantities rather than a single number (all on the 1,150-example test set, 55,200 query-heads). The positive-gap rate $\Pr(\Delta > 0)$ is 0.66. *Empirical compliance* $\Pr(m_F \leq \varepsilon)$—the fraction of heads already within budget—is 14.2%. *Analytic certified coverage*, where the closed-form bound $D_2^{\mathcal{F}} e^{-\Delta/\tau_0}$ itself falls at or below ε, is 2.85% (q=2); among empirically compliant heads the analytic certificate fires on 20% (its *efficiency*). The bound holds with *zero violations* on all positive-gap heads (q=2 and q=∞). Its *tightness*, the ratio of the analytic bound to the observed far mass, has median 4.2, so the bound is conservative by roughly a factor of four. Mean outside-evidence mass is 0.41 (median 0.36), concentrated in the upper layers (m_F by layer $0.21, 0.21, 0.53, 0.70$; certified fraction $6.4\%, 1.8\%, 1.7\%, 1.5\%$). Example-level stringent coverage (every analysed head of an example certified) is 0%: no single example is fully local. Stratifying by answer correctness, certified fraction is 3.5% on correctly answered vs 2.4% on incorrectly answered examples. The predicted answer position lies within the annotated support in 75% of examples, confirming the reader mostly reads from evidence even where individual heads do not certify.

Deletion analysis. Deleting the outside-evidence keys and renormalising, the analytic output-perturbation bound $2Bm_F$ (Note 11) holds with zero violations, but is loose: its median ratio to the observed displacement is 6.4. The certificate is therefore operationally *discriminative* rather than sharp—it separates populations without predicting each displacement. A certified head's readout moves by a mean relative 2.4% (median 1.5%) against 73% (median 46%) for uncertified heads (main-text operational Gibbs figure, panel c), so the certificate reliably identifies the sub-population on which deletion of outside-evidence keys stays within the stated budget.

Controls. Task annotation carries information beyond set size and beyond attention's own concentration, tested on the same reader with example-level bootstrap CIs (2,000 resamples). A size-matched *random* far set (fixed independently of the logits) certifies 0.52% (CI $[0.45, 0.59]$%); the task partition certifies 2.85% (CI $[2.61, 3.11]$%), a paired margin of 2.33 points (CI $[2.12, 2.56]$, excluding zero) so the task partition is statistically distinguishable from a random one of the same size. The *endogenous* top-decile-logit partition—where "far" is by construction where attention is already smallest, a method sanity check rather than independent evidence—certifies 21.0% (CI $[20.5, 21.5]$%). Both controls satisfy the bound with zero violations. The ordering random < task < endogenous confirms the annotation is informative without being tautological.

Training experiment. Because m_F is a differentiable function of the attention logits on the fixed annotation partition, it can be added to the loss. We fine-tune the same reader (adapter and span head trainable, ~ 0.9M adapter parameters injected in every layer via the `pf-hotpotqa` bottleneck configuration, backbone frozen) with $L = L_{\text{QA}} + \lambda \bar{m}_{\mathcal{F}}$, over three seeds, evaluating all 1,150 retained test examples; λ would be selected on the 287-example validation split. The reported certified fraction is analytic q=2 coverage. Relative to the matched λ=0 control (identical data, steps and seed; 4.2% ± 0.5 certified, 14% empirical compliance, exact-match 0.407 ± 0.012), the penalty raises analytic certified coverage to 15.6% (λ=0.5), 39.4% (λ=2) and 61.7% ± 0.9 (λ=8), and empirical compliance to 83%, pulling mean outside-evidence mass from 0.41 to 0.042 (below ε). The accuracy cost is small and quantified: exact-match falls to 0.389 ± 0.011 at λ=8 (paired per-seed drop 0.018, about two seed standard deviations) and is within noise at λ=2 (0.400 ± 0.013; token-F1 0.552 → 0.533). This is *evidence-supervised far-mass regularization followed by RCP certification*, not training the certificate or the D_2 bound directly (the objective penalizes the exact m_F); that 86%-class empirical compliance rather than 100% certification results is consistent with this distinction. It is a single-model, single-dataset proof of concept using gold supporting-fact supervision; the general claim is that the certificate is an operable training objective, not the specific coverage percentages. Script `hotpot_unified.py` is in the repository.

Cross-model and cross-dataset replication. To test whether the diagnosis and the trainability are properties of the certificate or of one adapter-tuned reader on one corpus, the entire pipeline—continued-trained λ=0 reader, full certificate audit, deletion coupling, task-vs-random control, and the λ=8 far-mass-regularised reader—was re-run over a 2×2 grid of two readers (BERT and RoBERTa, each with its `pf-hotpotqa` adapter) and two evidence-annotated multi-hop QA datasets (HotpotQA distractor and 2WikiMultiHopQA, a schema-matched parquet mirror with gold supporting-fact annotations), *three seeds per cell.* The BERT/HotpotQA cell reproduces the three-seed audit above (exact-match 0.407 ± 0.010, certified 4.2 ± 0.4%, task−random gap 2.28 ± 0.18), confirming the grid harness (Table S1). All four cells behave consistently across all three seeds: the Gibbs bound holds with *zero* violations at both λ=0 and λ=8 (and zero output-perturbation-bound violations) in every one of the twelve runs; the task partition certifies strictly more far-locality than a size-matched random partition, with the paired bootstrap margin excluding zero in every seed of every cell; and the far-mass penalty raises analytic certified coverage in every seed of every cell. The two architectures differ in an interpretable way: RoBERTa is the stronger reader (exact-match 0.51–0.52 vs 0.38–0.41) but its attention is measurably *more* non-local (mean outside-evidence mass 0.54–0.58 vs 0.41–0.43, positive-gap rate 0.49 vs 0.65), and the certificate reports this rather than hiding it—it still certifies a *larger* fraction of RoBERTa's positive-gap heads and shows a wider task-vs-random separation. The accuracy cost of the λ=8 penalty is 0.4–2.0 exact-match points on average (per-seed range 0.1–2.8). Scripts `replicate_grid.py` and `replicate_grid.json` are in the repository.

Limitations. The specific coverage percentages are dataset- and reader-dependent (Table S1); the general claims—the certificate holds with zero violations, the task partition beats a random one, and the far-mass penalty raises certified coverage—replicate across both readers and both datasets, but the exact numbers are not asserted to transfer beyond the four cells tested. A selection effect the bootstrap cannot address is the 21% retention filter: retained examples have shorter contexts (median 1,041 vs 1,215 tokens) and slightly more comparison questions (27% vs 19%), but the same evidence load (mean 2.35 vs 2.45 supporting sentences) and difficulty (all dev examples are labelled hard). The selection is mild and mechanistic—it favours contexts short enough to fit—and the reported percentages describe the retained subset, not the excluded longer-context majority (script `hotpot_retention.py`). Attention weights are not a causal explanation of the prediction; the certificate reports locality relative to a declared partition, not why the model answers as it does.

Part C · Note 11 — Output locality for normalised readouts

Table S1: Cross-model × cross-dataset replication of the task-grounded certificate. Each cell: canonical λ=0 continued-trained reader. "cert" is analytic q=2 certified coverage *conditional on positive gap* (i.e. over positive-gap query-heads only; the main text reports the unconditional coverage over *all* query-heads, which is this value times the positive-gap rate); "task−rand" is the paired margin (percentage points) between the task and size-matched random partitions, whose per-seed example-level bootstrap CI (2,000 resamples) excludes zero in every case; "cert λ:0→8" is certified coverage before and after the far-mass penalty; "ΔEM" is the mean exact-match cost of the penalty. Values are the mean over three seeds (± standard deviation); zero bound violations in all cells, all seeds, at both penalties.

Reader	Dataset	n	EM	mean $m_{\mathcal{F}}$	cert (λ=0)	task−rand	cert λ:0→8	ΔEM
BERT	HotpotQA	1150	0.407±0.010	0.41	4.2±0.4%	2.28±0.18	4.2→61.7±0.7%	1.80
BERT	2Wiki	960	0.375±0.006	0.43	3.4±0.5%	1.91±0.30	3.4→67.7±2.1%	1.25
RoBERTa	HotpotQA	1091	0.513±0.004	0.58	13.2±0.8%	5.91±0.22	13.2→38.6±5.7%	0.40
RoBERTa	2Wiki	960	0.518±0.011	0.54	17.6±2.2%	7.61±0.91	17.6→50.3±8.4%	2.01

Corollary S2 (Output locality for normalised readouts). Let $O = \sum_i p_i u_i$ be a normalised readout with bounded value vectors $\|u_i\| \le B$, and let $O_{\text{near}} = \sum_{i\notin\mathcal{F}}(p_i/(1-m_F))u_i$ be the output recomputed with the far set deleted and the softmax renormalised over the near set. Then at the certified temperature,

$$\|O - O_{\text{near}}\| \;\le\; 2B\, m_F \;\le\; 2B\varepsilon, \tag{12}$$

and for a probability-vector output the total-variation change is at most $m_F \le \varepsilon$. *Proof.* Writing $O - O_{\text{near}} = \sum_{i\notin\mathcal{F}} p_i(1-\frac{1}{1-m_F})u_i + \sum_{i\in\mathcal{F}} p_i u_i$, the first sum has total weight $(1-m_F)(\frac{1}{1-m_F}-1) = m_F$ and the second has total weight m_F; bounding each $\|u_i\| \le B$ gives $\|O - O_{\text{near}}\| \le 2Bm_F$, and $m_F \le \varepsilon$ by Theorem 2. ■ This gives the Gibbs certificate the same operational locality reading as the additive Corollary 2—deleting the far set moves the output by a certified amount—and is exactly the single-head (H=1) case of the multi-head bound (Note 12).

Part C · Note 12 — Full-layer attention bound

This note extends the bound from one head to the full attention layer (Proposition S4).

Setup. The Gibbs certificate of Note 10 controls a *single* head: at its certified temperature τ_{cert} (or, as reported here, at the deployed temperature), the output-relevant far weight for query q is the normalised far mass $m^h = \sum_{i\in\mathcal{F}} a_i^h$, where a_i^h are the softmax weights of head h and $\mathcal{F}$ is the far-key set. A transformer layer runs H heads in parallel and combines them: head h emits $o^h = \sum_i a_i^h \text{val}_i^h$ with value vectors $\text{val}_i^h = W_V^h x_i$, and the layer output is $O = W_O\,[\,o^1;\ldots;o^H\,]$ for an output projection W_O. We propagate the per-head guarantee to O.

Proposition S4 (Multi-head layer perturbation). Let O be the attention-layer output at query q and $\tilde{O}$ the output recomputed with the far-key set $\mathcal{F}$ deleted (softmax renormalised over the near keys). Then

$$\|O-\tilde{O}\|_2 \le \|W_O\|_2 \left(\sum_{h=1}^{H}\big(2\,m^h\,\|W_V^h\|_2\,X\big)^2\right)^{1/2}, \quad X = \max_i \|x_i\|_2, \tag{13}$$

with m^h the normalised far mass of head h (measured, or Gibbs-certified via Note 10). The bound is computable from the pretrained weights and the per-head far masses alone; no labels and no gradient are required.

Proof. Fix head h. Deleting $\mathcal{F}$ replaces $o^h = \sum_i a_i^h \text{val}_i^h$ by the near-only average $\tilde{o}^h = \sum_{i\notin\mathcal{F}}(a_i^h/n^h)\,\text{val}_i^h$ with $n^h = \sum_{i\notin\mathcal{F}} a_i^h = 1 - m^h$. Writing $\tilde{o}^h - o^h = \sum_{i\notin\mathcal{F}} a_i^h(\frac{1}{n^h}-1)\text{val}_i^h - \sum_{i\in\mathcal{F}} a_i^h \text{val}_i^h$ and bounding each value vector by

$V^h_{\max} := \max_i \|\mathrm{val}^h_i\|_2$, the first sum has total weight $n^h(\frac{1}{n^h} - 1) = m^h$ and the second has total weight m^h, so $\|o^h - \tilde{o}^h\|_2 \le 2m^h V^h_{\max}$. Since $\mathrm{val}^h_i = W^h_V x_i$, $V^h_{\max} \le \|W^h_V\|_2\, X$. Stacking the heads, $O - \tilde{O} = W_O\,[\, o^1 - \tilde{o}^1; \ldots; o^H - \tilde{o}^H\,]$, and $\|O - \tilde{O}\|_2 \le \|W_O\|_2\, \|[\, o^h - \tilde{o}^h\,]_h\|_2 = \|W_O\|_2 (\sum_h \|o^h - \tilde{o}^h\|_2^2)^{1/2}$. Substituting the per-head bound gives the statement. ■

Scope: bounded versus loose. The layer bound inherits the single-head scope. In the bounded-weight regime ($V_{\max} = 1$, as for prototypical networks) each $m^h \le \varepsilon$, and the layer perturbation is ε-small up to the fixed weight-norm factor. For attention the per-head far masses range over the measured 0.24–0.38 (per-head maximum across layers; the per-model means are 0.24–0.27, Note 8), so the bound is a genuine, computable perturbation guarantee but not ε-small: it certifies that the far field moves the layer output by a bounded amount, not that the amount is negligible. The bound is therefore a label-free, geometry-only far-field bound for the full attention layer—and it is loose by a measured factor, not vacuous.

Empirical confirmation (generic sanity check on a pretrained layer). This is a bound-validity check on an off-the-shelf pretrained layer, not part of the unified task-grounded pipeline of Note 10. On a real `bert-base-uncased` layer ($H = 12$ heads) over long contexts, we compute the bound from the pretrained W^h_V and W_O (spectral norms $\|W_O\|_2 \approx 2.4$–2.8, $\|W^h_V\|_2 \approx 1.1$–1.4) and the per-head far masses, and compare it to the *actual* $\|O - \tilde{O}\|_2$ from literally deleting the far keys and recomputing the layer output. Across four sampled layers $(0, 3, 6, 9)$ and all sampled queries the bound holds without exception. The actual far-deletion perturbation is a non-trivial 12–28% of the layer-output norm—attention genuinely moves under far-key deletion, as the unbounded weights predict—and the weight-norm bound exceeds it by a median factor of ≈ 47–$70\times$ (tightening to ≈ 12–$19\times$ if $V^h_{\max}$ is measured rather than bounded by $\|W^h_V\|_2 X$). The bound is thus valid and conservative; its looseness is the price of a purely weight-computable, geometry-only certificate (script and outputs in the repository).

Part D · Statistical validity

Part D · Note 13 — Conformal finite-sample query certificate

This note certifies a future query by conformal finite-sample coverage.

Statement. The mass $A = V_{\max} N_{\mathrm{eff}}$ is a query-average, so $\sigma^*(A)$ certifies the *mean* tail. A distribution-free guarantee for a *future single query* follows from conformal prediction with no labels. Let $A_1, \ldots, A_n$ be the per-query effective masses on an exchangeable calibration set of unlabelled queries, each measured at the conservative reference scale $A_i = V_{\max}\, N_{\mathrm{eff}}(y_i, \sigma_{\mathrm{pair}})$ so the calibration and the deployed frontier share one bandwidth. Let $A_{(1)} \le \cdots \le A_{(n)}$ be their order statistics. Fix a miscoverage level α and set $k = \lceil (n+1)(1-\alpha) \rceil$; this requires $k \le n$, i.e. $n \ge \lceil (1-\alpha)/\alpha \rceil$ (for $\alpha = 0.05$, $n \ge 19$), and otherwise $A_{\mathrm{conf}} := +\infty$ (no finite certificate). The resulting guarantee is finite-sample *marginal* coverage over the joint draw of calibration set and future query, not conditional coverage for a fixed realised calibration set.

Proposition S5 (Conformal query mass). Let $A_{\mathrm{conf}} = A_{(k)}$. For a fresh query Y drawn exchangeably with the calibration set, $\Pr[A_Y \le A_{\mathrm{conf}}] \ge 1 - \alpha$. Consequently deploying at $\sigma^*(A_{\mathrm{conf}}) = d/\sqrt{2\ln(A_{\mathrm{conf}}/\varepsilon)}$ certifies the far tail of the fresh query, $\mathrm{Tail}_{\sigma^*(A_{\mathrm{conf}})}(Y, d) \le \varepsilon$, with probability at least $1 - \alpha$.

Proof. By exchangeability of $A_1, \ldots, A_n, A_Y$, the rank of A_Y among the $n+1$ values is uniform on $\{1, \ldots, n+1\}$, so $\Pr[A_Y \le A_{(k)}] \ge k/(n+1) \ge 1 - \alpha$ for $k = \lceil (n+1)(1-\alpha) \rceil$. On the event $A_Y \le A_{\mathrm{conf}}$, deploying at $\sigma^*(A_{\mathrm{conf}}) \le \sigma_{\mathrm{pair}}$ and using that N_{eff} is nondecreasing in σ (Proposition 2, so $A_Y(\sigma^*) \le A_Y(\sigma_{\mathrm{pair}}) = A_Y$), the aggregate bound gives $\mathrm{Tail}_{\sigma^*}(Y,d) \le A_Y(\sigma^*)K_{\sigma^*}(d^2) \le A_Y K_{\sigma^*}(d^2) \le A_{\mathrm{conf}} K_{\sigma^*(A_{\mathrm{conf}})}(d^2) = \varepsilon$ (the frontier is defined so the last equality holds), which occurs with probability at least $1-\alpha$. $\square$

Empirical confirmation. On the three CIFAR-100 backbones, splitting the held-out queries into an $n = 7000$-point calibration set and a disjoint 3000-query validation set at $\alpha = 0.05$, the fresh-query coverage $\Pr[A_Y \le A_{\mathrm{conf}}]$ is 0.947, 0.943, and 0.953; with 3000 validation queries the binomial standard error on a coverage of 0.95 is ≈ 0.004, so all three are within one to two standard errors of the 0.95 target, as the marginal guarantee predicts. Further, A_{conf} exceeds the mean mass A by 10–17%, so the conformal bandwidth is slightly smaller (more conservative) than the mean-certified one; zero validation queries exceed the budget at $\sigma^*(A_{\mathrm{conf}})$. The conformal wrapper thus upgrades the mean certificate to a distribution-free, label-free, finite-sample guarantee for the next query, at the cost of a fraction of the certified bandwidth (script `renyi_conformal.py` in the repository).

Backbone	Method	tail/ε	giant%	acc%
ResNet-18 (512d)	naive	5068	100	45.2
	median heur.	215695	100	39.6
	k-NN(7)	120953	100	40.5
	σ^*	0.08	7	56.6
	grid (labels)	0.01	4	57.2
	Silverman	125520	100	40.4
ResNet-50 (2048d)	naive	4892	97	33.9
	median heur.	210717	100	25.3
	k-NN(7)	129605	100	26.6
	σ^*	0.09	7	51.0
	grid (labels)	0.00	0	56.1
	Silverman	127973	100	26.6
ViT-B/16 (768d)	naive	9963	100	65.7
	median heur.	216464	100	63.1
	k-NN(7)	110309	100	63.7
	σ^*	0.16	4	72.5
	grid (labels)	0.00	1	73.0
	Silverman	123721	100	63.6

Table S2: Full comparison on the three frozen CIFAR-100 backbones. σ^* is the only label-free rule shown that holds the interference budget while approaching labelled grid-search accuracy.

Part E · The empirical program

Part E · Note 14 — Cross-modal calibration

This note collects the empirical frontier: the label-free bandwidth against labelled and heuristic baselines across modalities. The certificate's central empirical claim is that a single label-free bandwidth holds the interference budget while matching labelled grid-search accuracy across modalities. Table S2 reports the three frozen CIFAR-100 backbones; Table S3 the text and tabular corpora.

Corpus (dim)	method	tail/ε	acc%
DBpedia-14	σ^*	0.24	92.1 ±0.6
text (384)	grid (lab.)	—	92.3 ±0.5
	median	45609	89.0 ±0.7
	k-NN(7)	33492	89.1 ±0.6
AG-News-4	σ^*	0.27	87.9 ±1.1
text (384)	grid (lab.)	—	88.0 ±1.1
	median	13018	85.7 ±0.8
	k-NN(7)	10272	85.8 ±0.9
Yahoo-10	σ^*	0.30	66.9 ±0.5
text (384)	grid (lab.)	—	67.8 ±0.3
	median	32631	65.8 ±0.4
	k-NN(7)	24822	66.0 ±0.4
Covertype-7	σ^*	0.10	65.0 ±0.8
tab. (54)	grid (lab.)	—	70.8 ±1.1
	median	20755	53.9 ±1.5
	k-NN(7)	102	60.0 ±1.6

Table S3: Full cross-modal results over five seeds. The naive bandwidth, omitted for space, has tail/ε from 3.6×10^2 to 2.3×10^3.

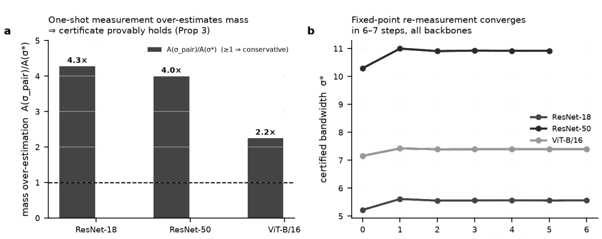

Figure S3: **The measured mass is conservative.** (a) Measuring A once at σ_{pair} over-estimates the mass at σ^* by 2.2–4.3× on all three backbones. (b) The certified bandwidth used in deployment is the one-shot value; the exact self-consistent bandwidth is the root of the monotone envelope, found by bisection (the naive re-measurement iteration is shown for reference only, and is not relied on for convergence).

Part E · Note 15 — Hard truncation

This note compares the smooth certified field against a hard-truncated compact-support operator with the same locality objective.

Hard-truncation comparator. A same-objective locality baseline replaces the smooth kernel by its hard-truncated version $K_\sigma^{\text{trunc}}(r^2) = K_\sigma(r^2)\mathbf{1}\{r \le d\}$, which has *exact* zero far mass by construction rather than a calibrated ε-bounded tail. Selecting its bandwidth by the same labelled oracle grid on the ResNet-18 field, the truncated field reaches accuracy 0.558 against the smooth σ^* field's 0.566 (a loss of 0.8 points), and its hard decisions agree with the smooth full field on only 89.0% of queries. The comparison isolates the value of calibrating a smooth, globally supported field rather than replacing it with a discontinuous compact-support operator: exact locality is available at the price of a small accuracy loss, an 11% decision disagreement with the operator actually deployed, and the loss of continuity under query perturbation. RCP instead keeps the smooth field intact and certifies that its far interference is $\le \varepsilon$, so the certified and deployed readouts are the same object.

Part F · Verification and reproducibility

Part F · Note 16 — Numerical verification

Purpose. The construction rests on two facts: the participation-ratio tail bound (Proposition 1) and the monotonicity of N_{eff} in σ that makes the one-shot measurement conservative (Proposition 2). Because a certificate is only as good as these, we verify them directly on random configurations, independently of the deep-network experiments.

Tail bound (Proposition 1). Over 20,000 random all-far configurations—each with a radius d, a far-set size $|\mathcal{F}|$ drawn in $[5, 200]$, radii $r_i > d$, and a bandwidth σ spanning a wide range of d—we evaluate the worst-case aligned tail $\sum_{i\in\mathcal{F}} \omega_i$ (all $|v_i| = V_{\max}$) against the certificate $\mathrm{cert}(A, \sigma, d) = N_{\mathrm{eff}}\, K_\sigma(d^2)$. The ratio never exceeds one (maximum 0.98, mean 0.76; zero violations), and Fig. S4a shows *where* the tightening lives: it approaches one only as $N_{\mathrm{eff}}/|\mathcal{F}| \to 1$ (near-uniform weights), and is strictly below the union/count bound $|\mathcal{F}|\, K_\sigma(d^2)$ whenever the far weights are unequal—so the participation ratio is not merely valid but strictly tighter than the count it replaces.

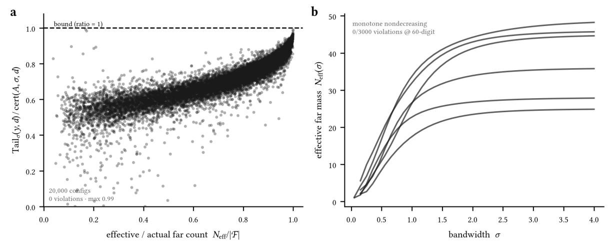

Figure S4: **Independent verification of the two load-bearing results.** (a) Tail-bound tightness over 20,000 random far configurations: the ratio stays below one everywhere (max 0.98, zero violations), nearing one only as $N_{\mathrm{eff}}/|\mathcal{F}| \to 1$. (b) $N_{\mathrm{eff}}(\sigma)$ is monotone nondecreasing (Proposition 2), zero violations at 60-digit precision over 3,000 trials.

Conservativeness (Proposition 2). Over 3,000 random far-sets evaluated on a 40-point σ grid at 60-digit precision, $N_{\mathrm{eff}}(\sigma)$ is monotone nondecreasing with zero violations (smallest consecutive increment 2.1×10^{-49}, i.e. numerically non-negative); Fig. S4b shows representative curves. Apparent violations at ordinary floating precision are underflow artifacts as $\sigma \to 0$ and vanish at high precision.

A shorter route to Proposition 2. The monotonicity has a one-line reading worth recording. The far weights $\omega_i = \exp(-r_i^2/2\sigma^2)$ are a tempered family with inverse temperature $\beta = 1/2\sigma^2$, and $N_{\mathrm{eff}} = (\sum_i \omega_i)^2 / \sum_i \omega_i^2$ is exactly the inverse-Simpson (Rényi-2) diversity of the normalised weights. Diversity indices are nondecreasing as a distribution is tempered toward uniformity (increasing σ, decreasing β, flattens the weights), which is Proposition 2. The explicit differencing proof of Note 3 is self-contained; this identity connects it to the diversity-index literature and makes the direction of the monotonicity immediate.

Empirical confirmation of the Gibbs and Rényi bounds. *The following checks, folded here from the normalised-certificate note, confirm Theorem 2 on real fields.*

Empirical confirmation on real attention (endogenous diagnostic). This paragraph uses the *endogenous* top-decile-logit partition—“near” is the keys the field itself ranks highest—so it measures attention’s self-concentration, not locality relative to external evidence (for which see the task-grounded HotpotQA verification of Note 10). On the real query/key vectors of BERT-base and GPT-2 over long contexts (~ 360 keys/head), with the near anchor the top-decile-logit keys and $\mathcal{F}$ the remainder, the bound $m_F \le D_q^{\mathcal{F}} e^{-\Delta/\tau}$ holds with *zero violations* across $\sim$33,000 query-heads per model at both $q = 2$ and $q = \infty$. Two readouts are available, and we are explicit about which we report. The *operational* readout evaluates the field at its *actual* deployed temperature $\tau_0 = \sqrt{d_k} = 8$: the measured normalised far mass is 0.27 (BERT) and 0.24 (GPT-2)—a bounded, computable far-field weight at the temperature the model really runs, not ε-small. The *diagnostic* readout is the per-query certified temperature $\tau_{\mathrm{cert},y} = \Delta_y / \ln(D_{q,y}(\tau_0)/\varepsilon)$, computed head-by-head from each query’s own logit gap and Hill number measured at τ_0; here every query has $\tau_{\mathrm{cert},y} \ll \tau_0$ (median ≈ 0.02), so this is pure cooling and the one-shot certificate applies to each query separately—at its own $\tau_{\mathrm{cert},y}$ the far mass drops below ε (to $< 10^{-10}$) with zero violations across all $\sim 66{,}000$ query-heads. This is a query-adaptive diagnostic, not a single deployable transformer temperature: it says how much more peaked each head would have to be to be provably local, and the answer ($8 \to 0.02$) is that certifying attention this way would radically change the model. The operative statement for a *pretrained* transformer is therefore the inverse one—report the far mass (or, equivalently, the certified gap $\Delta_{\text{cert}} = \tau_0 \ln(D_q^{\mathcal{F}}/\varepsilon)$) at the fixed deployed τ_0—which is what the 0.24–0.27 figures give (script `gibbs_attention.py` in the repository).

Empirical confirmation on a bounded-weight readout. The prototypical case exercises the same certificate with a squared-distance energy $E_k = \|f(x) - c_k\|^2$, anchor $E_0 = r^2_{\text{near}}$ and gap $\Delta = d^2_{\text{far}} - r^2_{\text{near}}$. On a trained conv-4 ProtoNet (Omniglot, 89.8% held-out accuracy over 7,200 certified queries), the pointwise bound $m_F \le D_q^{\mathcal{F}}(\tau)e^{-\Delta/\tau}$ holds at both $q = 2$ and $q = \infty$ (at the reference temperature $\tau_0 = 1$ the measured far mass is 0.080 against a mean bound of 0.165 at q=2; zero violations of the bound). The point of interest is the *executable* one-shot certificate. This field runs near budget, so unlike attention it does not have room to cool freely: for about 30% of queries the naive $\tau_{\text{OS}} = \Delta/\ln(D_{q,0}/\varepsilon)$ exceeds τ_0 (it would ask the field to *warm*, which raises D_q and fails), and the mean τ_{OS} is close to τ_0. Deploying at the corrected $\tau_{\text{cert}} = \min(\tau_0, \tau_{\text{OS}})$ resolves this cleanly: cooling queries are certified because cooling only sharpens the far weights (lowering $D_q^{\mathcal{F}}$ below the reference $D_{q,0}$), warming queries already satisfy the bound at τ_0, and the measured certified far mass has mean 0.015 (q=2) and 0.017 (q=∞) and *maximum* 0.0476 $< \varepsilon$ *over all* 7,200 *queries at both* q, with *zero* certificate violations. This is the direct Gibbs analogue of the additive one-shot theorem, and it is exactly why the certified temperature must be $\min(\tau_0, \tau_{\text{OS}})$ and not the naive τ_{OS}: measuring D_q at one reference temperature and inverting can call for warming, which the monotone bound does not license. The contrast with attention is instructive: a transformer runs far warmer than budget, so $\tau_{\text{OS}} \ll \tau_0$ everywhere (pure cooling) and the exponential collapses the far mass to near zero; a ProtoNet runs near budget, so the min is load-bearing on a third of queries. In both regimes the one-shot certificate holds with zero violations (script `gibbs_proto.py` in the repository).

Non-Gaussian replication. The conservativeness and frontier also hold for non-Gaussian log-concave tails; the Laplace case is shown below.

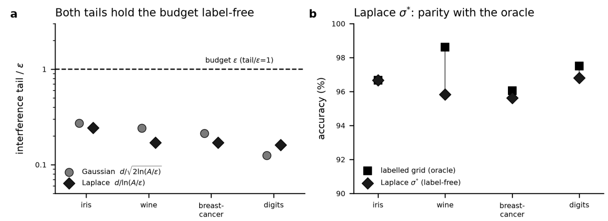

Figure S5: **The calibration is not Gaussian-specific.** On four public datasets, the Laplace frontier $\sigma^* = d/\ln(A/\varepsilon)$ holds the interference budget label-free and approaches a labelled grid search anchored to the Laplace scale.

Part F · Note 17 — Robustness

Is the near/far radius meaningful? Assumption (A2) asks that the radius d separate "intended" from "interfering" sources. We test this directly by measuring, at σ^*, the kernel-weighted fraction of near-set mass that shares the query's class—the quantity the vote actually uses. It is 0.31, 0.25, and 0.47 on ResNet-18, ResNet-50, and ViT-B/16, against a chance level of 0.01 for 100 classes: a 25–47× enrichment, so the near set is strongly class-aligned. Two qualifications. This is a *plurality*, not a majority—on two of three backbones most individual near sources are other-class—so "intended sources" should be read as "near sources," and the decision-equality guarantee of Theorem 1 is equality with the near-field *vote*, which is itself only weakly pure: the certificate guarantees *which* sources decide, not that they decide correctly. Accuracy is a separate, measured outcome (Table S2), not a corollary of the certificate.

Sensitivity to the budget and radius. The budget ε and the radius percentile are inputs, so we sweep them: $\varepsilon \in \{0.01, 0.05, 0.1, 0.2\}$ crossed with the $\{5, 10, 20\}$th distance percentile for d. Deployed accuracy varies by only 2.2 points (ResNet-18), 6.7 (ResNet-50), and 2.0 (ViT) across the whole grid. This flatness must be read correctly: over that grid the realised σ^* itself spans only a 1.34–1.41× range (in concentrated high-dimensional geometry the distance percentiles are close and $\sqrt{2\ln(A/\varepsilon)}$ moves slowly), so the informative statement is the *conjunction*—across the grid, accuracy is flat *and* the certified tail stays below budget, while a bandwidth only 4–6× larger (the distance heuristics) fails catastrophically. The budget is a genuine dial with a mild, monotone effect on the operating point, not a hidden accuracy knob.

Pointwise versus average. The certificate is derived per query, but A is measured as an *average* N_{eff} over the deployment queries, so the reported tail/ε is strictly an average-case quantity; a per-query guarantee requires the tail to hold at *every* query, not on average. We therefore measure the full per-query distribution on the ResNet-18 field (10,000 held-out queries at σ^*): tail/ε has mean 0.08, 95th percentile 0.18, 99th percentile 0.20, and *maximum* 0.24—not a single query exceeds the budget. The average-A certificate is thus empirically pointwise here, with fourfold headroom at the worst query.

An a priori certificate by quantile choice. The average is a convenience, not a requirement: the same closed form delivers a strict per-query certificate if A is chosen as an upper quantile of N_{eff} rather than its mean. This is not empirical luck but a direct reading of the frontier. Because $\sigma^*(A)$ is decreasing in A and $\mathrm{Tail}_\sigma(y, d) \le V_{\max} N_{\text{eff}}(y)\, K_\sigma(d^2)$ pointwise, deploying at $\sigma^*(A_q)$ with A_q the $(1-\alpha)$-quantile of N_{eff} over the region guarantees the budget at *every* query whose $N_{\text{eff}}(y) \le A_q$—that is, at all but at most an α-fraction of the region by construction; taking $A_q = \sup_y N_{\text{eff}}(y)$ ($\alpha = 0$) certifies the whole region uniformly. The quantile level is thus a dial on the allowed per-query failure rate, read off the same geometry, at no labels. Empirically the tail is far below this worst case: at $\alpha = 0.01$ (the 99th percentile) the *measured* maximum tail/ε over 10,000 queries stays inside the budget on all three backbones—0.19 on ResNet-18, 0.17 on ResNet-50, 0.27 on ViT-B/16, with zero exceedances anywhere—while accuracy is unchanged to within 0.2 points because $\sigma^*(A_q)$ is only about one percent smaller than $\sigma^*(A_{\text{mean}})$. Even the fully uniform choice $A_{\max} = \sup_y N_{\text{eff}}(y)$ costs essentially nothing: accuracy changes by at most 0.21 points from the mean-A field on all three backbones (marginally higher, in fact), again with zero exceedances by construction. These are empirical maxima and quantiles over the observed query sets, not a priori bounds on an unseen query; a finite-sample guarantee for a future exchangeable query is provided only by the conformal construction of Note 13. With that caveat, the per-query certificate is inexpensive to deploy across the suite, with the quantile level set to the failure rate a deployment will tolerate.

Drift of the radius d. The conservativeness result (Proposition 2) is stated at fixed near/far radius d. Because d is the tenth percentile of pairwise centre distances, computed here from a fixed 4,000-centre subsample, its value depends on the field only weakly: across a 40× increase in field size the re-estimated radius changed by approximately 1%, while the dominant change in the certificate was the effective far mass. Re-estimating d per stage therefore corrects only a small term, and this is all the growing-memory argument requires. We make no claim about the estimator's asymptotic rate: pairwise centre distances are dependent, so the sample-quantile $O_p(n^{-1/2})$ heuristic does not apply directly, and we rely on the measured 1% stability rather than an asymptotic argument.

Frozen-geometry stress test. Corollary S1 states the certificate depends on the geometry only through fixed dissimilarities. We test this on three controlled settings (Fig. S6). *(i) Anisotropic families.* A two-class problem separable on one axis is embedded with an uninformative axis inflated by ratio 1–8. Isotropic-Euclidean RCP holds the budget throughout (median tail/$\varepsilon \approx 0.24$, 0 violations) but its decision accuracy falls from 1.00 to 0.94 as the noise axis dominates the Euclidean distance; a frozen global Mahalanobis metric that undoes the inflation recovers accuracy to 0.98–1.00 while holding the same budget. *(ii) Two-moons.* With Euclidean proximity differing from intrinsic connectivity, both an isotropic-Euclidean field and a frozen out-of-sample graph-geodesic field hold the budget (tail/$\varepsilon = 0.28$ and 0.30, 0 violations) and agree on every held-out decision. *(iii) Densification.* Across $n = 200$–3200 at fixed axis ratio, both geometries keep tail/$\varepsilon < 1$ with 0 violations while σ^* shrinks with density. The certificate is thus a property of the frozen geometry, not of the Euclidean metric specifically: a better geometry improves the usefulness of the near/far split without changing the guarantee. The isotropic field is not claimed to recover manifold structure; the point is that RCP certifies whatever locality the frozen geometry encodes.

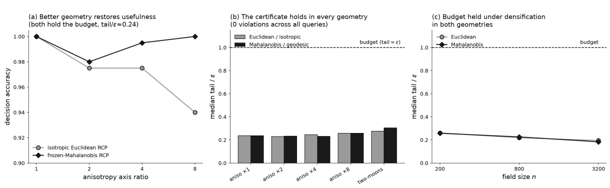

Figure S6: **The certificate transfers across frozen geometries.** (a) As the anisotropy axis ratio grows, isotropic-Euclidean RCP loses decision accuracy while a frozen-Mahalanobis metric recovers it; both hold the interference budget. (b) Median tail/ε stays below the budget in every geometry and configuration, with zero per-query violations. (c) The budget is held under 16× densification in both the Euclidean and frozen-geometry variants.

Part F · Note 18 — Reproducibility

Computational specification. All reported quantities are computed as follows. Features are standardised per dimension (z-score using the training mean and standard deviation); distances are Euclidean in that standardised space. The radius d is the 10th percentile of pairwise centre distances over a 4,000-centre subsample; A and N_{eff} are the far-mask participation ratio averaged over 2,000 held-out queries measured at σ_{pair}; the reported tail/ε is the *average* far mass over held-out queries divided by ε (the full per-query distribution, including the maximum, is reported for the flagship field above); the giant-component percentage is the largest connected component as a fraction of 5,000 sampled centres, with an edge whenever two centres lie within 3σ (this diagnostic is *illustrative* of field locality, not itself the certificate—a larger σ mechanically connects more centres, so it visualises the collapse rather than measuring the interference bound). The labelled comparator ("grid") selects the bandwidth that maximises held-out accuracy over a 30-point log-spaced grid; it is a grid search on labelled accuracy, not a k-fold procedure. The grid spans $[0.5, 30]$ for the vision fields and $[0.3, 40]$ for the text and modality fields; for the lower-dimensional public datasets it is log-spaced to cover the data scale of each dataset. The accompanying result files for the vision, text, and modality experiments record the exact grid and selected bandwidth for each. The interference budget is $\varepsilon = 0.05$ throughout. Every run uses a fixed seed; the five-seed re-samples vary only the drawn centres and queries. Scripts and result files accompany the manuscript.

Growing-field experiments. The re-certification result (main text, "Re-calibration follows memory density") uses two growth protocols on CIFAR-100 penultimate features. *Density growth* (three backbones: ResNet-18, ResNet-50, ViT-B/16) densifies the field from 3 to 120 examples per class over five stages (300 to 12,000 centres, all 100 classes present throughout so the test set is fixed); a labelled grid search fixes σ once on the sparse 3-per-class field, and at each stage we report σ^* (re-measured), that frozen bandwidth, and a per-stage re-tuned oracle grid. *Class-incremental growth* (ResNet-18) instead adds classes over five stages ($10 \to 25 \to 50 \to 75 \to 100$) at a fixed 120 examples per class, restricting the test set to classes seen so far (standard class-incremental protocol); the labelled grid is fixed once on the initial 10-class field. All bandwidth selection, radius, mass, and tail definitions match the computational specification above ($\varepsilon = 0.05$, d at the 10th distance percentile, A over 3,000 held-out queries). The class-incremental run is a reported *null*: adding classes at fixed density leaves the field geometry essentially unchanged (the radius d drifts by -3.5% over the full range and σ^* shrinks only from 6.26 to 5.43), so the frozen bandwidth stays within budget (tail/ε from 0.01 to 0.23, never violating) and ties σ^* on accuracy—confirming that the re-certification advantage is driven by densification, not by class count. Both scripts and their result files accompany the manuscript.

Backbone	Stage	σ^* tail/ε	median tail/ε	k-NN tail/ε
ResNet-18	10 cls	0.13	2.1×10^4	1.3×10^4
	100 cls	0.09	2.1×10^5	1.2×10^5
ResNet-50	10 cls	0.11	2.1×10^4	1.4×10^4
	100 cls	0.09	2.1×10^5	1.3×10^5
ViT-B/16	10 cls	0.14	2.3×10^4	1.3×10^4
	100 cls	0.15	2.2×10^5	1.1×10^5

Table S4: Label-free median and k-NN scales, re-computed at each density stage, still overshoot the interference budget by four to five orders of magnitude, while σ^* (re-measured at each stage) holds tail/$\varepsilon \approx 0.1$. The label-free heuristics' accuracy falls from 0.82/0.83 (10 cls) to 0.40/0.41 (100 cls) on ResNet-18 and comparably on the other backbones, against certified σ^* accuracies $0.888 \to 0.564$ (R18), $0.833 \to 0.512$ (R50), $0.957 \to 0.720$ (ViT). Script `grow_baselines.py` in the repository.

Re-computed label-free baselines under densification. To test adaptation fairly, we also recompute the median and k-nearest-neighbour ($k = 7$) bandwidth scales *label-free at every density stage*, so that these heuristics receive the updated geometry exactly as σ^* does. They still miss the interference budget by a wide margin at every stage (Table S4): median and k-NN drive tail/ε to 2×10^4–2×10^5 across the sweep while losing between about 1 and 26 accuracy points relative to σ^* depending on backbone and density (largest on ResNet-50, smallest on the sparse ViT field), whereas re-measured σ^* holds tail/$\varepsilon \approx 0.1$ throughout. Adapting the *scale* to the geometry is therefore not the same as holding the *interference budget*: only the certificate is constructed to do the latter. The labelled-grid optima ($\sigma = 4.9$, 6.6, 8.0 for the three backbones) lie in the interior of the 0.5–30 search grid, so they are not grid-boundary artefacts; where a labelled optimum fell within one grid step of an endpoint we expanded the grid and re-ran.